Use case narratives:

Use Case 1: Create the input file

Primary actor: Researcher

Secondary actors: text editor (External software)

Goal: Define the input parameters to be used in running the simulation.
Preconditions:

- Researcher has opened a black document in a text editor

Use:

- Researcher enters the following parameters into a text file, in a pre-defined format:
    - Placement of initial bacteria
    - Bacteria motility force
    - Bacteria reproduction rate
    - Initial nutrient concentration
    - EPS production rate
    - Nutrient threshold for EPS production
    - EPS size
    - EPS-cell attraction range
    - Coefficients of friction between particles
    - Simulation boundary
    - Maximum agent limit
    - Intermediate output interval: to adjust the framerate for later visualization

Alternative paths:

No alternative paths exist at this stage, however any errors or changes in this interaction will result in prompts from the system in later interactions.

Postconditions:

- Input file exists
- Input file has the correct parameters and format

Use Case 2: Run Biofilm Simulation

Primary actor: Researcher

Secondary actors: Simulation module and User interface

Goal: Run a simulated model of early-stage biofilm growth using defined input parameters.
Preconditions:

- Simulation parameters have been entered into an input file
- Simulation engine is operational.

Use:

- Researcher launches the simulation module.
- Researcher enters the name of the input file.
- Researcher Runs the Simulation.
- Simulation module:
    - Validates inputs
    - Runs calculations to simulate bacterial growth
    - Outputs simulation data to a new file
- Simulation completes

Alternative paths:

- Validate inputs: If input file is not found, system alerts user and allows them to try again – redirect to the initial input screen.
- Validate inputs: If input file contains invalid data, system alerts user and allows them to try again – redirect to the initial input screen.
- Runs calculations: If parameters become too large, system stops the simulation and alerts user – data up to this point is saved to the output file.
    - User is directed to the options menu (as if the simulation ended normally)

Postconditions:

- Simulation has run to completion
- Output file has been created and populated

Use Case 3: View Simulation

Primary actor: Researcher

Secondary actors: Visualisation module and User interface

Goal: Visualise the simulation that has just completed.
Preconditions:

- Simulation has been run and an output file produced.
- User has received the output file (or name thereof)

- Simulation engine is operational.

Use:

- User selects visualisation parameters
- User selects to View the simulation
- Visualization module processes output into:
    - Video-like simulation over time (faster than true time)
    - Images generated at set time points from the simulation
- User is directed back to the options menu

Alternative paths:

- Different visualisation options will lead to different outputs.
- User may exit the visualisation at any time
- Selecting visualisation parameters: any errors in the parameters will lead to a prompt from the system, instructing the user on how to correct the inputs before running the visualization.

Postconditions:

- Visualisation has completed

Use Case 4: Generate simulation graphs

Primary actor: Researcher

Secondary actors: Graphing module and User interface

Goal: Visualise the simulation data in the form of various graphs
Preconditions:

- Simulation has been run and an output file produced.
- User has received the output file (or name thereof)
- Simulation engine is operational.

Use:

- User selects graphing parameters
- User selects to View graphs
- Graphing module processes output into various graphs based on user selection
- User can export graphs to a file

Alternative paths:

- Different graphing options will lead to different output graphs
- User may exit the graphing section at any time

- Selecting graphing parameters: any errors in the parameters will lead to a prompt from the system, instructing the user on how to correct the inputs before the graphs may be generated.

Postconditions:

- Graphs have been viewed
- File of graphs has been created (if user selected this option)